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Fastening apparatus

Abstract

A fastening apparatus includes a base, a clasp body, and an elastic body. The base is configured to be arranged at a first object. The clasp body is movably disposed at the base, and movably couples or separates the second object. The elastic body is disposed between the base and the clasp body and pushes against the clasp body, such that the clasp body is normally located at an engaging connection position or normally located at a disengagement position. Thus, the base can be assembled with the first object and can be engaging connected with or disengaged from the second object using the clasp body, so as to complete quick coupling and separation of two objects, further achieving effects of repeated quick coupling and separation.

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Patent No.	Application Date	Country	CPC
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I736410	12/2020	TW	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This non-provisional application claims priority under 35 U.S.C. § 119(a) on Patent Application No(s). 112100347 filed in Taiwan, R.O.C. on Jan. 5, 2023, the entire contents of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present disclosure provides a fastening apparatus, and in particular to a fastening apparatus for repeatedly and quickly coupling and separating at least two objects.

2. Description of the Related Art

(3) In general, when at least one object is to be coupled, lock connection is usually performed using a screw for coupling the object.

(4) In the conventional fixing means above, although at least one object can be fixed and coupled in a form that is not easily separated, in addition to unhandy assembly, the least one object may not be easily removed due to the fixed coupling means using the screw.

BRIEF SUMMARY OF THE INVENTION

(5) In view of the prior art above, the present disclosure provides a fastening apparatus in aim of achieving effects of repeated quick coupling and easy separation.

(6) To achieve the above and other objects, the present disclosure provides a fastening apparatus including a base, a clasp body, and an elastic body. The base is configured to be arranged at a first object. The clasp body is movably disposed at the base, and movably couples to or separates from a second object. The elastic body is disposed between the base and the clasp body and pushes against

the clasp body, such that the clasp body is normally located at an engaging connection position or normally located at a disengagement position.

(7) In the fastening apparatus, a bottom of the base is provided with at least one assembly section, which is for the base to be arranged at the first object.

(8) In the fastening apparatus, the assembly section is arranged at the first object by means of an engaging connection, welding connection, lock connection, expansion connection, or adhesion.

(9) In the fastening apparatus, the bottom of the base is provided with at least one anti-rotation assembly section, which is for the base to be arranged at the first object and provides an anti-rotation function.

(10) In the fastening apparatus, the base is provided with at least one locking section, which can coordinate with a locking fixture for the base to be arranged at the first object.

(11) In the fastening apparatus, the base is provided with a containment area, in which the clasp body and the elastic body are movably disposed.

(12) In the fastening apparatus, each of two sides of the clasp body is provided with a sliding portion, and each of two sides in the base is provided with a sliding groove, wherein each of the sliding portions is docked with each of the sliding grooves so as to movably dispose the clasp body at the base.

(13) In the fastening apparatus, each of the sliding portions is provided with a limiting slider, and each of the sliding grooves is provided with a sliding restriction section, wherein each of the limiting slider is disposed at each of the sliding restriction section, so as to limit a movement distance of the clasp body on the base.

(14) In the fastening apparatus, each of two sides of the base is provided with a guiding section, and each of the limiting sliders is disposed at each of the sliding restriction sections via each of the guiding sections.

(15) In the fastening apparatus, one end of the clasp body is provided with a first limiting flange or a second limiting flange. The first limiting flange or the second limiting flange can be used to engaging connect and couple the second object, or disengage and separate the second object.

(16) In the fastening apparatus, one end of the first limiting flange is provided with a guiding slope section, and the second object pushes against the guiding slope section, such that the clasp body is pushed and reset by the elastic body after moving horizontally, and the first limiting flange or the second limiting flange engaging connects and couples with the second object.

(17) In the fastening apparatus, the second object is provided with a recessed section, and the recessed section of the second object is fitted into the second limiting flange via the first limiting flange, such that the first limiting flange abuts against the second object.

(18) In the fastening apparatus, one end of the base is provided with a contact section which corresponds to the first limiting flange, such that the first limiting flange abuts against a top surface of the second object and the contact section abuts against a bottom surface of the second object.

(19) In the fastening apparatus, the clasp body is provided with an operating section, which is operable to control the clasp body to move horizontally on the base, such that the clasp body coordinates with the base for engaging connection or disengagement, further coupling or separating the second object.

(20) In the fastening apparatus, one end of the elastic body is integrally connected with the clasp body, and the other end of the elastic body pushes against the base, such that the elastic body pushes the clasp body to be normally located at a lock position or normally located at an unlock position.

(21) In the fastening apparatus, one end of the elastic body is provided with a compression segment, and the other end of the elastic body is provided with an assembly section. The compression segment and the clasp body are integrally connected, and the assembly section is joined at a securement section of the base, such that the elastic body pushes the clasp body to be normally located at a lock position or normally located at an unlock position.

- (22) In the fastening apparatus, the elastic body is an elastic element bending continuously.
- (23) In the fastening apparatus, the elastic element is a plastic member, a metal member, or an elastic piece integrally connected with the clasp body.
- (24) In the fastening apparatus above, an appearance of the elastic element or an appearance of an elastic region generated by the elastic element is in an S shape, a continuous S shape, an approximate S shape, or an approximate continuous S shape.
- (25) Thus, in the fastening apparatus of the present disclosure, the base can be assembled with the first object and can be engaging connected with or disengaged from the second object using the clasp body, so as to complete quick coupling and separation of two objects, further achieving objects of repeated quick coupling and separation.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of an appearance of a fastening apparatus of an embodiment of the present disclosure.
- (2) FIG. 2 is a schematic diagram of an appearance of a fastening apparatus of an embodiment of the present disclosure.
- (3) FIG. 3 is an exploded schematic diagram of a fastening apparatus of an embodiment of the present disclosure.
- (4) FIG. 4 is an exploded schematic diagram of a fastening apparatus of an embodiment of the present disclosure.
- (5) FIG. 5 is a schematic diagram of a cross-section state of a fastening apparatus of an embodiment of the present disclosure.
- (6) FIG. 6 is a schematic diagram of a state of use of a fastening apparatus of an embodiment of the present disclosure.
- (7) FIG. 7 is a schematic diagram of a state of use of a fastening apparatus of an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

- (8) To facilitate understanding of the object, characteristics, and effects of the present disclosure, embodiments together with the attached drawings for the detailed description of the present disclosure are provided.
- (9) Refer to FIG. 1 to FIG. 7. As shown in the drawings, the present disclosure provides a fastening apparatus **1** including a base **11**, a clasp body **12**, and an elastic body **13**.
- (10) The base **11** can be arranged at a first object **10**.
- (11) The clasp body **12** is movably disposed at the base **11**, and horizontally moves and coordinates with the base **11** for engaging connection or disengagement, such that the clasp body **12** couples or separates a second object **20**.
- (12) The elastic body **13** is disposed between the base **11** and the clasp body **12** and pushes against the clasp body **12**, such that the clasp body **12** is normally located at an engaging connection position or normally located at a disengagement position.
- (13) During use, the fastening apparatus **1** can be assembled on the first object **10**, and then the second object **20** is directly pressed down upon one end of the clasp body **12** (or by manually applying a force upon the clasp body **12**), such that the clasp body **12** moves horizontally and compresses the elastic body **13**. Once the second object **20** abuts against a top of one end of the base **11**, the elastic body **13** is reset and the clasp body **12** is pushed, and a bottom of one end of the clasp body **12** coordinates with the top of one end of the base **11** so as to be engaging connected to the second object **20**, such that the clasp body **12** is locked at the second object **20**, hence completing coupling between the clasp body **12** and the second object **20** (as shown in FIG. 6 and

FIG. 7).

(14) To separate the second object **20**, the clasp body **12** is moved horizontally in a direction away from the second object **20** and the elastic body **13** is compressed, such that the clasp body **12** separates from the second object **20**. Then, the elastic body **13** is reset and the clasp body **12** is pushed to be ready to perform a next operation of the clasp body **12**. Thus, the base **11** can be assembled with the first object **11**, and the base **11** can be engaging connected and disengaged with the coordination of the clasp body **12**, so as to couple or separate the clasp body **12** from the second object **20** to complete coupling and separation of two objects, hence achieving the effects of repeated quick coupling and separation.

(15) In addition to the embodiments above, in an embodiment of the present disclosure, the base **11** is provided with at least one assembly section **111**. The assembly section **111** is disposed at a bottom of the base **11**, and is for the base **11** to be arranged at the first object **10** by means of a soldering connection, welding connection, engaging connection, lock connection, expansion connection, or adhesion. Thus, the base **11** can be stably arranged at the first object **10** to facilitate a movement operation of the clasp body **12**.

(16) In addition to the embodiments above, in an embodiment of the present disclosure, the base **11** is provided with at least one anti-rotation assembly section **112**. The anti-rotation assembly section **112** is disposed at the bottom of the base **12** and near the assembly section **111**. The assembly section **111** and the anti-rotation assembly section **112** are for the base **11** to be arranged at the first object **10**, and the anti-rotation assembly section **112** provides an anti-rotate function once the base **11** is arranged at the first object **10**. Thus, the base **11** can be stably arranged at the first object **10** to facilitate a movement operation of the clasp body **12**.

(17) In addition to the embodiments above, in an embodiment of the present disclosure, the base **11** is further provided with at least one locking section **113**. The locking section **113** is for coordinating with a locking fixture **101**, so that in addition to be arranged at the first object **10** by using the assembly section **111** and the anti-rotation assembly section **112**, the base **11** can be assembled and connected to the first object **10** by simultaneously (or separately) using the locking section **113** in coordination with the locking fixture **101**. Thus, the base **11** can be stably arranged at the first object **10** to facilitate a movement operation of the clasp body **12**.

(18) In addition to the embodiments above, in an embodiment of the present disclosure, the base **11** is provided with a containment area **114**, in which the clasp body **12** and the elastic body **13** are movably disposed. Thus, when the clasp body **12** moves horizontally on the base **11** to compress or release the elastic body **13**, the containment area **114** can be used as a movement space, thereby achieving an effect of smooth movement to facilitate a lock or unlock operation.

(19) In addition to the embodiments above, in an embodiment of the present disclosure, each of two sides of the clasp body **12** is provided with a sliding portion **121**, and each of two sides in the base **11** is provided with a sliding groove **115**, wherein each of the sliding portions **121** is docked with each of the sliding groove **114**, so as to movably dispose the clasp body **12** at the base **11**. When the clasp body **12** moves horizontally on the base **11**, the containment area **114** can be used as a movement space, and the sliding portions **12** in coordination with the sliding grooves **115** can serve as guide and travel distance limitation for movement, thereby achieving an effect of smooth movement to facilitate a lock or unlock operation.

(20) In addition to the embodiments above, in an embodiment of the present disclosure, each of the sliding portions **121** is provided with a limiting slider **122**, and each of the sliding grooves **115** is provided with a sliding restriction section **116**, wherein each of the limiting sliders **122** is disposed in each of the sliding restriction sections **116**, so as to limit a movement distance of the clasp body **12** on the base **11**. Each of two sides of the base **11** is provided with a guiding section **117**, and each of the limiting sliders **122** is disposed at each of the sliding restriction sections **116** via each of the guiding sections **117**. When the clasp body **12** moves horizontally on the base **11**, the containment area **114** can be used as a movement space, and the coordination of the sliding portions **121**, the

limiting sliders **122**, the sliding grooves **115**, and the sliding restriction sections **116** is used as guide and travel distance limitation for the movement of the clasp body **12** on the base **11**, thereby achieving an effect of smooth movement to facilitate a lock or unlock operation.

(21) In addition to the embodiments above, in an embodiment of the present disclosure, one end of the clasp body **12** is provided with a first limiting flange **123** and a second limiting flange **124**. The first limiting flange **123** and the second limiting flange **124** can be used to engaging connect and couple the second object **20**, or disengage and separate the second object **20**. One end of the first limiting flange **123** is provided with a guiding slope section **125**, and the second object **20** pushes against the guiding slope section **125**, such that the clasp body **12** is pushed and reset by the elastic body **13** after moving horizontally, and the first limiting flange **123** and the second limiting flange **124** engaging connect and couple the second object **20**. The second object **20** is provided with a recessed section **201**, and the recessed section **201** of the second object **20** is fitted into the second limiting flange **124** via the first limiting flange **123**, such that the first limiting flange **123** abuts against the second object **20**. Moreover, one end of the base **11** is provided with a contact section **118** which corresponds to the first limiting flange **123**, such that the first limiting flange **123** abuts against a top surface of the second object **20** and the contact section **118** abuts against a bottom surface of the second object **20**.

(22) On the basis of the embodiments above, during use, the fastening apparatus **1** can be assembled on the first object **10**, and then the recessed section **201** of the second object **20** is directly pressed down upon the guiding slope section **125** of the clasp body **12**, such that the clasp body **12** moves horizontally and compresses the elastic body **13**. Once the recessed section **201** of the second object **20** is fitted into the second limiting flange **124** via the first limiting flange **123**, the elastic body **13** is reset and the clasp body **12** is reset, so as to abut the first limiting flange **123** against the top surface of the second object **20** and abut the contact section **118** against the bottom surface of the second object **20**, hence completing coupling between the clasp body **12** and the second object **20** (as shown in FIG. 6 and FIG. 7).

(23) To separate the second object **20**, the clasp body **12** is moved horizontally in a direction away from the second object **20** and the elastic body **13** is compressed, such that the first limiting flange **123** and the second limiting flange **124** separate from the recessed section **201** of the second object **20**. Then, the elastic body **13** is reset and the clasp body **12** is pushed to facilitate a next operation of the clasp body **12**. Thus, the base **11** can be assembled with the first object **11**, and the base **11** can be engaging connected and disengaged with the coordination of the clasp body **12**, so as to couple or separate the clasp body **12** from the second object **20** to complete coupling and separation of two objects, hence achieving the effects of repeated quick coupling and separation.

(24) In addition to the embodiments above, in an embodiment of the present disclosure, the base **12** is provided with an operating section **126**, which is operable to control the movement of the clasp body **12** on the base. Thus, when the clasp body **12** is moved, a force is applied upon the operating section **126** for a control operation, so as to better control the clasp body **12** to coordinate with the base **11** for engaging connection or disengagement, further coupling or separating the second object **20** to complete coupling and separation of two objects, hence achieving the effects of repeated quick coupling and separation.

(25) In addition to the embodiments above, in an embodiment of the present disclosure, one end of the elastic body **13** is integrally connected with the clasp body **12** and the other end of the elastic body **13** pushes against the base, such that the elastic body **13** pushes the clasp body **12** to be normally located at a lock position or normally located at an unlock position. In an embodiment of the present disclosure, one end of the elastic body **13** is provided with a compression segment **131**, and the other end of the elastic body **13** is provided with an assembly section **132**. The compression segment **131** and the clasp body **12** are integrally connected, and the assembly section **132** is joined at a securement section **119** of the base **11**, such that the elastic body **13** pushes the clasp body **12** to be normally located at a lock position or normally located at an unlock position.

(26) On the basis of the embodiments above, when the clasp body **12** moves horizontally, the coordination of the compression segment **131** and the assembly section **132** is employed to first make the clasp body **12** compress the elastic body **13**, and then the elastic body **13** is reset and the clasp body **12** is pushed, such that the elastic body **13** pushes the clasp body **12** to be normally located at the lock position or normally located at the unlock position, thereby allowing the clasp body **12** to perform engaging connection or disengagement. Thus, the clasp body **12** is coupled to or separated from the second object **20**, so as to complete coupling and separation of two objects, hence achieving the effects of repeated quick coupling and separation.

(27) In addition to the embodiments above, in an embodiment of the present disclosure, the elastic body **13** can be an elastic element bending continuously. The elastic body **13** is a plastic member, a metal member, or an elastic piece integrally connected with the clasp body **12**. An appearance of the elastic body **13** or an appearance of an elastic region generated by the elastic body **13** is in an S shape, a continuous S shape, an approximate S shape, or an approximate continuous S shape. Thus, the elastic body **13** can similarly be made to push the clasp body **12** to be normally located at the lock position or normally located at the unlock position, thereby allowing the clasp body **12** to perform engaging connection or disengagement. Thus, the clasp body **12** is coupled to or separated from the second object **20**, so as to complete coupling and separation of two objects, hence achieving the effects of repeated quick coupling and separation.

(28) The present disclosure is described by way of the preferred embodiments above. A person skilled in the art should understand that, these embodiments are merely for describing the present disclosure are not to be construed as limitations to the scope of the present disclosure. It should be noted that all equivalent changes, replacements and substitutions made to the embodiments are to be encompassed within the scope of the present disclosure. Therefore, the scope of protection of the present disclosure should be accorded with the broadest interpretation of the appended claims.

Claims

1. A fastening apparatus, comprising: a base, configured to be arranged at a first object; a clasp body, movably disposed at the base, the clasp body being moved so as to couple or separate a second object; and an elastic body, disposed between the base and the clasp body and pushing against the clasp body, such that the clasp body is normally located at an engaging connection position or normally located at a disengagement position, wherein one end of the elastic body is integrally formed with the clasp body, and one other end of the elastic body pushes against the base, such that the elastic body pushes the clasp body to be normally located at a lock position or normally located at an unlock position.
2. The fastening apparatus according to claim 1, wherein a bottom of the base is provided with at least one assembly section, the assembly section is for the base to be arranged at the first object, or the assembly section is arranged at the first object by means of engaging connection, solder connection, welding connection, lock connection, expansion connection, or adhesion.
3. The fastening apparatus according to claim 1, wherein a bottom of the base is provided with at least one anti-rotation assembly section, which is for the base to be arranged at the first object and provides an anti-rotation function.
4. The fastening apparatus according to claim 1, wherein the base is provided with at least one locking section, which coordinates with a locking fixture for the base to be arranged at the first object.
5. The fastening apparatus according to claim 1, wherein the base is provided with a containment area, in which the clasp body and the elastic body are movably disposed.
6. The fastening apparatus according to claim 1, wherein each of two sides of the clasp body is provided with a sliding portion, each of two sides in the base is provided with a sliding groove, and each of the sliding portions is docked with each of the sliding grooves such that the clasp body is

movably disposed at the base; or each of the sliding portions is provided with a limiting slider, each of the sliding grooves is provided with a sliding restriction section, and each of the limiting sliders is disposed at each of the sliding restriction sections so as to limit a movement distance of the clasp body on the base; or each of two sides of the base is provided with a guiding section, and each of the limiting sliders is disposed at each of the sliding restriction sections via each of the guiding sections.

7. The fastening apparatus according to claim 1, wherein one end of the clasp body is provided with a first limiting flange or a second limiting flange, and the first limiting flange and the second limiting flange are for engaging connecting and coupling the second object, or disengaging and separating the second object; or one end of the first limiting flange is provided with a guiding slope section, and the second object pushes against the guiding slope section, such that the clasp body is pushed and reset by the elastic body after moving horizontally, and the first limiting flange and the second limiting flange engaging connect and couple the second object; or the second object is provided with a recessed section, and the recessed section of the second object is fitted into the second limiting flange via the first limiting flange, such that the first limiting flange abuts against the second object; or one end of the base is provided with a contact section which corresponds to the first limiting flange, such that the first limiting flange abuts against a top surface of the second object and the contact section abuts against a bottom surface of the second object.

8. The fastening apparatus according to claim 1, wherein the clasp body is provided with an operating section, which is operable to control the clasp body to move horizontally on the base, such that the clasp body coordinates with the base for engaging connection or disengagement, further coupling or separating the second object.

9. The fastening apparatus according to claim 1, wherein one end of the elastic body is provided with a compression segment, one other end of the elastic body is provided with an assembly section, the compression segment and the clasp body are integrally connected, and the assembly section is joined at a securement section of the base, such that the elastic body pushes the clasp body to be normally located at a lock position or normally located at an unlock position.

10. The fastening apparatus according to claim 1, wherein the elastic body is an elastic element bending continuously; or the elastic element is a plastic member, a metal member, or an elastic piece integrally connected with the clasp body; or an appearance of the elastic element or an appearance of an elastic region generated by the elastic element is in an S shape, a continuous S shape, an approximate S shape, or an approximate continuous S shape.

11. A fastening apparatus, comprising: a base, configured to be arranged at a first object; a clasp body, movably disposed at the base, the clasp body being moved so as to couple or separate a second object; and an elastic body, disposed between the base and the clasp body and pushing against the clasp body, such that the clasp body is normally located at an engaging connection position or normally located at a disengagement position, wherein each of two sides of the clasp body is provided with a limiting slider, each of two sides of the base is provided with a sliding restriction section, each of the limiting sliders is disposed at each of the sliding restriction sections so as to limit a movement distance of the clasp body on the base.
